

# Jaegun Song

Ann Arbor, MI · US citizen

[jaeguntsong@gmail.com](mailto:jaeguntsong@gmail.com) · [github.com/Astral1119](https://github.com/Astral1119) · [linkedin.com/in/jaeguns](https://linkedin.com/in/jaeguns) · [jaegun.dev](https://jaegun.dev)

New-grad software engineer working on language tooling, reactive evaluation, and agent infrastructure, with a background in spreadsheet calculation engines. Shipped production agent systems at an AI tutoring startup.

## Education

University of Michigan, Ann Arbor

B.S. in Computer Science (Expected May 2027) · GPA 3.74

## Experience

### Agent Engineering Intern

May 2026 – Aug 2026

*Aristotle (AI tutoring startup, employee #6)*

- Shipped 111 merged PRs in three months across the tutor agent, observability, evals, email, and billing; most commits on the team in my final month
- Built live session observability on Cloudflare Durable Objects; traced dropped events to instance eviction (every 10–60s), fixed with epoch-tagged sequences
- Shipped PDF-by-link ingestion (credential-free fetcher, SSRF guards, atomic quota admission), replacing up to 155k tokens per turn with on-demand page reads
- Built a session-review agent in production that critiques ended sessions in Slack and proposes prompt edits behind a human approval gate

### Data Science Intern

May 2025 – Aug 2025

*Enveda Biosciences / True Ventures Fellow*

- Coauthored [a paper](#) on NMR molecular networking for structure characterization (*Communications Chemistry*, 2025)
- Built a Discord–Slack bridge for a 190+ member AI founders network

### Programmer

Sep 2024 – May 2025

*University of Michigan School of Information*

- Developed a plugin integrating U-M Maizey into Google Sheets
- Migrated legacy research dashboards to Fly.io, reducing deployment time and dependency overhead

### Intern

May 2024 – Aug 2024

*U-M Information & Technology Services (Emerging Technologies Group)*

- Built [gotocollege.umich.edu](https://gotocollege.umich.edu), matching students with 5M+ scholarships using agentic AI retrieval
- Designed and deployed RAG utilities that improved Maizey document recall from 74% → 86%

## Projects

### Lattice — Rust, Language Design

In progress. Functional language unifying spreadsheet formulas and structured programming, with one type system and lazy reactive evaluation across source files and cells.

### Formulary — TypeScript

Package manager for Google Sheets: dependency resolution, versioned named-function packages, reproducible builds.

### astral.lol — Google Sheets, Language Design

Research playground exploring spreadsheets as a programming language: esolang interpreters (Brainfuck, Befunge), pathfinding, fuzzy lookup, and stateful computation using only formulas and iterative calculation (no Apps Script).

- Found and reported Google Sheets crash bugs by email; most were patched
- Built a chess engine entirely in spreadsheet formulas (regex move generation, neural-network evaluation, depth-2 search); measured at 1308 Elo against Stockfish (95% CI 1260–1354, 320 games)

### Brickwise — Python, Genetic Algorithms, Mesh Processing

Converted 3D meshes into buildable LEGO brick layouts with a genetic algorithm. *3rd Place, Silicon Flatirons 2025.*

### Cartularium — Open Source, Technical Writing

Lead maintainer of a spreadsheet community's open-source projects, including [sheets.wiki](https://sheets.wiki) (527 pages; 464 Discord citations).

## Skills

**Languages:** Python, Rust, C++, TypeScript, Node.js, SQL

**Frameworks/Tools:** Cloudflare Workers, LLM and agent tooling, PyTorch, scikit-learn, PostgreSQL, React

**Other:** Google Sheets (advanced), Michigan Seal of Biliteracy (Spanish)

## Community

- Advent of Code 2023, 2025 (Google Sheets):** Ranked #1 among spreadsheet-based solvers; formulas only
- Community Administration:** Administrator of a 4,000+ member spreadsheet-development help community; organized online chess tournaments featuring grandmasters